Energy-Efficient Hybrid STAR-RIS via Sparse Active Amplification and Gain Control

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***Abstract*—Hybrid active-passive simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RISs) can improve spectral efficiency (SE), yet their energy efficiency (EE) is constrained by amplifier noise and hardware power. This letter characterizes the EE-SE boundary through a multi-component power model covering BS transmit power, controller/bias power, switching overhead, and gain-dependent amplification power. An amplification on–off rule, a closed-form SE-oriented common-gain law, and a power-minimizing gain refinement are embedded into a finite-grid sparse-activation solver for ES, TS, and SS protocols. Simulations under Nakagami-m fading show that sparse amplification improves moderate-to-high-SE operation by balancing BS-power reduction against active-noise and hardware overhead.**

***Index Terms*—STAR-RIS, active RIS, hybrid active-passive RIS, energy efficiency, spectral efficiency, Pareto boundary, amplifier noise.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) can enhance wireless propagation by passive beam manipulation [1]. STAR-RISs extend this capability to simultaneous transmission and reflection, enabling a common surface to serve users on both sides [2], [3]. This property is attractive for asymmetric two-side coverage, but purely passive STAR-RIS remains limited by multiplicative path loss.

Embedding sparse active elements can mitigate this loss, but the resulting gain is paid for through amplifier noise and hardware power [4]–[7]. Measurement-driven studies further indicate that RIS power includes controller/bias and switching terms rather than a single constant [8], [9]. In STAR-RIS, these effects are more pronounced because the two sides jointly compete for aperture, power split, and active hardware budget.

Existing active or hybrid STAR-RIS studies have addressed related scenarios, including NOMA and SWIPT [10], [11]. Nevertheless, the joint effect of amplifier noise, sparse activation, dynamic switching overhead, and side-wise EE-SE tradeoff has not been characterized under a common boundary-generation framework. To address this gap, this letter makes four specific contributions. First, it formulates a multi-component STAR-RIS power model that separates BS PA power, controller/bias consumption, reconfiguration overhead, and gain-dependent amplification power. Second, it derives a side-wise gain structure that exposes when sparse amplification is SINR-beneficial and how the common gain should be refined for EE. Third, it embeds these rules into a finite-grid sparse solver with cost-aware ranking. Finally, it gives a unified ES/TS/SS boundary comparison and extracts design rules.

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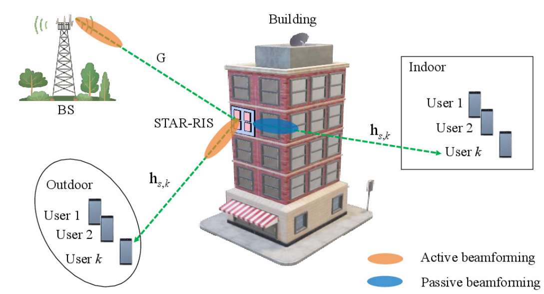

Fig. 1. Hybrid active-passive STAR-RIS-assisted two-side downlink.

II. System model

A. Hybrid active-passive STAR-RIS downlink

Consider the downlink illustrated in Fig. 1, where a BS with M antennas serves K_t transmission-side users and K_r reflection-side users through an N-element STAR-RIS. The user-index sets are $\mathcal{K}_t = \{1, \ldots, K_t\}$ and $\mathcal{K}_r = \{1, \ldots, K_r\}$. Under the ES protocol [3], the nth STAR-RIS element uses

$$\theta_{n,s} = \sqrt{\rho_{n,s}}\beta_{n,s}e^{j\phi_{n,s}}, \qquad s \in \{t, r\}, \tag{1}$$

where $\rho_{n,s}$ is the power-splitting coefficient, $\beta_{n,s}$ is the amplification factor, and $\phi_{n,s}$ is the phase shift. The coefficients satisfy $\rho_{n,t} + \rho_{n,r} = 1$, $0 \le \rho_{n,s} \le 1$, and $1 \le \beta_{n,s} \le \beta_{\max}$. Passive elements use $\beta_{n,s} = 1$, while active elements use $\beta_{n,s} > 1$ on their assigned side. Let $\mathcal{A}_t$ and $\mathcal{A}_r$ denote the active-element sets for transmission and reflection. Each active element is assigned to at most one side, so $\mathcal{A}_t \cap \mathcal{A}_r = \emptyset$, and $\mathcal{A} = \mathcal{A}_t \cup \mathcal{A}_r$.

Denote by $\mathbf{G} \in \mathbb{C}^{N\times M}$ the BS-to-STAR-RIS channel, and by $\mathbf{h}_{s,k} \in \mathbb{C}^N$ the STAR-RIS-to-user channel on side s for user k. The received signal at user (s, k) is

$$y_{s,k} = \mathbf{h}_{s,k}^H \mathbf{\Theta}_s \mathbf{G} \sum_{s'\in\{t,r\}} \sum_{j\in\mathcal{K}_{s'}} \mathbf{w}_{s',j} x_{s',j} + \mathbf{h}_{s,k}^H \mathbf{n}_{a,s} + n_{s,k}, \tag{2}$$

where $\mathbf{\Theta}_s = \operatorname{diag}(\theta_{1,s}, \ldots, \theta_{N,s})$, $\mathbf{w}_{s,j} \in \mathbb{C}^M$ is the BS beamformer for user (s, j), $x_{s,j}$ is the unit-power information symbol, and $n_{s,k} \sim \mathcal{CN}(0, \sigma_0^2)$ is thermal noise. The vector $\mathbf{n}_{a,s}$ collects the amplified noise contributions from the active elements on side s. Following the active-RIS noise model in [4], [7], the active-noise vector is modeled as

$$\mathbf{n}_{a,s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_{a,s}), \qquad \mathbf{C}_{a,s} = \operatorname{diag}(c_{1,s}, \ldots, c_{N,s}), \tag{3}$$

where $c_{n,s} = \rho_{n,s}\beta_{n,s}^2\sigma_a^2$ for $n \in \mathcal{A}_s$ and $c_{n,s} = 0$ otherwise. Here, σ_a^2 is the input-referred active-element noise variance, and $\mathbf{n}_{a,s}$ denotes the equivalent amplified noise contribution.

The SINR of user (s,k) becomes

$$\gamma_{s,k} = \frac{\left|\mathbf{h}_{s,k}^{H}\mathbf{\Theta}_{s}\mathbf{G}\mathbf{w}_{s,k}\right|^{2}}{\sum_{\substack{s'\in\{t,r\},\ j\in\mathcal{K}_{s'}\\(s',j)\neq(s,k)}}\left|\mathbf{h}_{s,k}^{H}\mathbf{\Theta}_{s}\mathbf{G}\mathbf{w}_{s',j}\right|^{2}+\sigma_{a,s,k}^{2}+\sigma_{0}^{2}}, \tag{4}$$

where $\sigma_{a,s,k}^{2}=\mathbf{h}_{s,k}^{H}\mathbf{C}_{a,s}\mathbf{h}_{s,k}$ is the active-noise term seen at user (s,k).

B. Power-consumption model

Following measurement-based RIS power models [8], [9], the consumed power is

$$\begin{aligned} P_{\mathrm{tot}} = &\frac{1}{\eta_{\mathrm{PA}}}\sum_{s\in\{t,r\}}\sum_{k\in\mathcal{K}_{s}}\|\mathbf{w}_{s,k}\|^{2}+P_{\mathrm{ctrl}}+P_{\mathrm{bias}}|\mathcal{A}| \\ &+f_{u}(E_{\mathrm{sw},0}+E_{\mathrm{sw},1}|\mathcal{A}|) \\ &+\sum_{n\in\mathcal{A}_{t}}\mu_{t}(\beta_{n,t}^{2}-1)+\sum_{n\in\mathcal{A}_{r}}\mu_{r}(\beta_{n,r}^{2}-1), \end{aligned} \tag{5}$$

where η_{PA} is the BS PA efficiency, $|\mathcal{A}|$ is the number of active elements, P_{ctrl} is the controller power, P_{bias} is the per-active-element bias power, f_u is the update frequency, $E_{\mathrm{sw},0}$ and $E_{\mathrm{sw},1}$ are the base and per-active-element switching energies per update, and μ_t,μ_r are gain-dependent amplifier-power coefficients. The four terms in (5) represent the PA-normalized BS transmit power, static controller/bias power, update-rate-weighted switching power, and gain-dependent amplification power, respectively.

C. Equivalent scalar model

To obtain a tractable EE-SE boundary, the full model is reduced to an equivalent scalar form using four standard approximations: residual multiuser leakage in (4) is neglected, common side-wise splitting and gain are used with $\rho_{n,s}=\rho_s$ and $\beta_{n,s}=\beta_s$ for $n\in\mathcal{A}_s$, dominant cascaded paths are phase-aligned, and the side target R_s is enforced through equal per-user targets R_s/K_s.

Let $\mathbf{v}_{s,k}$ be a normalized post-ZF BS direction. The scalar BS-to-element gain is represented by the stream-averaged effective gain

$$|g_n| \triangleq \frac{1}{K_t+K_r}\sum_{s\in\{t,r\}}\sum_{k\in\mathcal{K}_s}|[\mathbf{G}\mathbf{v}_{s,k}]_n|.$$

With $h_{s,k,n}$ denoting the nth entry of $\mathbf{h}_{s,k}$, define

$$u_{s,k,n}=|g_n|\,|h_{s,k,n}|. \tag{6}$$

Then the side-wise effective coherent term for user (s,k) is

$$\psi_{s,k}(\beta_s)=\sum_{n\in\mathcal{P}_s}\sqrt{\rho_s}u_{s,k,n}+\beta_s\sum_{n\in\mathcal{A}_s}\sqrt{\rho_s}u_{s,k,n}, \tag{7}$$

where $\mathcal{P}_s=\{1,\ldots,N\}\setminus\mathcal{A}_s$ contains passive elements and those active only for the opposite side. The corresponding active-noise term is

$$\zeta_{s,k}(\beta_s)=\sigma_0^2+\rho_s\beta_s^2\sigma_a^2\sum_{n\in\mathcal{A}_s}|h_{s,k,n}|^2. \tag{8}$$

Under the equal per-user target R_s/K_s, the minimum side power for meeting side target R_s is

$$p_s^{\min}(\beta_s)=\sum_{k\in\mathcal{K}_s}\frac{(2^{R_s/K_s}-1)\zeta_{s,k}(\beta_s)}{\psi_{s,k}^2(\beta_s)}. \tag{9}$$

III. Problem formulation

The sum spectral efficiency and normalized energy efficiency are

$$R_{\Sigma}=\sum_{s\in\{t,r\}}\sum_{k\in\mathcal{K}_s}\log_2(1+\gamma_{s,k}), \tag{10}$$

$$\mathrm{EE}_{\mathrm{norm}}=\frac{R_\Sigma}{P_{\mathrm{tot}}}\quad \text{(bit/J/Hz)}. \tag{11}$$

For bandwidth B, the physical EE equals $BR_\Sigma/P_{\mathrm{tot}}$ in bit/J. The normalized metric in (11) is used throughout. Let $\mathcal{X}\triangleq\{\mathbf{W},\rho_t,\rho_r,\boldsymbol{\phi},\beta_t,\beta_r,\mathcal{A}_t,\mathcal{A}_r\}$. The corresponding normalized-EE maximization problem is

$$\begin{aligned} \max_{\mathcal{X}}\quad & \frac{R_\Sigma}{P_{\mathrm{tot}}} \\ \text{s.t.}\quad & \log_2(1+\gamma_{s,k})\geq\bar{R}_{s,k},\ \forall s,k, \\ & \sum_{s\in\{t,r\}}\sum_{k\in\mathcal{K}_s}\|\mathbf{w}_{s,k}\|^2\leq P_{\max}^{\mathrm{BS}}, \\ & 1\leq\beta_s\leq\beta_{\max},\quad s\in\{t,r\}, \\ & \mathcal{A}_t\cap\mathcal{A}_r=\emptyset,\quad \mathcal{A}=\mathcal{A}_t\cup\mathcal{A}_r, \\ & \rho_t+\rho_r=1,\quad 0\leq\rho_s\leq1,\quad s\in\{t,r\}, \\ & \rho_s\beta_s^2\sigma_a^2\leq\Gamma_s,\quad s\in\{t,r\}. \end{aligned} \tag{12}$$

Here, Γ_s is an optional side-wise noise-injection or hardware-safety limit. Problem (12) is non-convex because its fractional objective couples beamforming, STAR coefficients, common gains, and active-set selection.

To trace an EE-SE Pareto-boundary approximation, the ϵ-constraint formulation is adopted

$$\begin{aligned} \mathcal{P}(\epsilon):\quad \min_{\mathcal{X}}\quad & P_{\mathrm{tot}} \\ \text{s.t.}\quad & R_\Sigma\geq\epsilon, \\ & \text{constraints of (12).} \end{aligned} \tag{13}$$

In the solver, the generic QoS terms are instantiated as $\bar{R}_{s,k}=R_s/K_s$ with $R_t=\alpha\epsilon$ and $R_r=(1-\alpha)\epsilon$, so the per-user constraints realize the side targets in (9). Since the solver uses structured sparse activation and finite grids, the generated curve is an achievable Pareto-boundary approximation rather than a certified global optimum. The model structure is then exploited to derive transferable design rules.

IV. Gain structure

A. Amplification on–off criterion

Given a side support, the key decision is whether the common active gain should be switched on, i.e., whether increasing β_s above one improves the side-wise SINR. Define

$$\begin{aligned} a_s &= \min_k\sum_{n\in\mathcal{P}_s}\sqrt{\rho_s}u_{s,k,n}, \\ b_s &= \min_k\sum_{n\in\mathcal{A}_s}\sqrt{\rho_s}u_{s,k,n}, \\ c_s &= \sigma_0^2, \\ d_s &= \max_k\rho_s\sigma_a^2\sum_{n\in\mathcal{A}_s}|h_{s,k,n}|^2. \end{aligned} \tag{14}$$

Here, a_s and b_s are conservative worst-user coherent-gain aggregates from passive and active contributions, while c_s and d_s are the thermal-noise baseline and active-noise coefficient. Let $\bar{p}_s$ denote an auxiliary normalized side power used only for this local monotonicity analysis. The worst-user SINR lower bound is

$$\underline{\gamma}_s(\beta_s)=\frac{\bar{p}_s(a_s+b_s\beta_s)^2}{c_s+d_s\beta_s^2}. \tag{15}$$

The coherent term grows linearly with β_s, whereas the injected active noise grows quadratically. Differentiating (15) gives

$$\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s} = \frac{2\bar{p}_s(a_s + b_s\beta_s)(b_s c_s - a_s d_s \beta_s)}{(c_s + d_s\beta_s^2)^2}. \tag{16}$$

Thus, at the passive-gain point $\beta_s = 1$, amplification is locally useful if

$$\left.\frac{\mathrm{d}\underline{\gamma}_s}{\mathrm{d}\beta_s}\right|_{\beta_s=1} > 0 \iff b_s c_s > a_s d_s. \tag{17}$$

Equation (17) gives the amplification on–off criterion: the active coherent-gain aggregate b_s, weighted by the thermal-noise baseline c_s, must dominate the passive coherent-gain aggregate a_s weighted by the active-noise coefficient d_s. When this inequality holds, amplification is locally beneficial for the SINR lower bound. Otherwise, the amplifier should remain off in this local test. Bias, switching, and amplification-power costs further tighten the final activation decision.

Proposition 1. *For the regular hybrid active-passive case $a_s > 0$, $b_s > 0$, $c_s > 0$, and $d_s > 0$, $\underline{\gamma}_s(\beta_s)$ is unimodal on $[1, \beta_{\max}]$. Its SE-oriented gain is*

$$\beta_{s,\mathrm{SE}}^{\star} = \left[\frac{b_s c_s}{a_s d_s}\right]_{[1,\beta_{\max}]}, \tag{18}$$

where $[x]_{[l,u]} = \min\{u, \max\{l, x\}\}$. The non-regular cases follow from the same derivative behavior. If $b_s = 0$, set $\beta_{s,\mathrm{SE}}^{\star} = 1$. If $a_s = 0$ with $b_s > 0$, or if $d_s = 0$ with $b_s > 0$, $\underline{\gamma}_s(\beta_s)$ is nondecreasing and $\beta_{s,\mathrm{SE}}^{\star} = \beta_{\max}$. If $|\mathcal{A}_s| = 0$, set $\beta_s = 1$.

Proof: In the regular case, the sign of (16) is determined only by $b_s c_s - a_s d_s \beta_s$, which changes sign once. Hence (15) is unimodal and the stationary point is $b_s c_s/(a_s d_s)$, clipped to $[1, \beta_{\max}]$. The boundary cases follow directly from (16): $b_s = 0$ makes the derivative non-positive, while $a_s = 0$ with $b_s > 0$ or $d_s = 0$ with $b_s > 0$ makes (15) nondecreasing. ■

B. Power-minimizing gain

The gain in (18) is SE-oriented. EE optimization must also account for BS transmit power and active-amplifier power. For a target side rate R_s, define the side-wise cost $\Psi_s(\beta_s)$ as the sum of the transmit-power term induced by (9) under the worst-user coefficients and the gain-dependent active-amplifier power:

$$\Psi_s(\beta_s) = \frac{K_s(2^{R_s/K_s} - 1)(c_s + d_s\beta_s^2)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^2} + \mu_s|\mathcal{A}_s|(\beta_s^2 - 1), \tag{19}$$

$$\Psi_s'(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)(a_s d_s \beta_s - b_s c_s)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^3} + 2\mu_s|\mathcal{A}_s|\beta_s, \tag{20}$$

$$\Psi_s''(\beta_s) = \frac{2K_s(2^{R_s/K_s} - 1)}{\eta_{\mathrm{PA}}(a_s + b_s\beta_s)^4}\left(a_s^2 d_s - 2a_s b_s d_s \beta_s + 3b_s^2 c_s\right) + 2\mu_s|\mathcal{A}_s|. \tag{21}$$

Proposition 2. *Assume the hybrid active-passive case $a_s > 0$, $b_s > 0$, and $d_s > 0$. Under the condition*

$$\beta_{\max} \le \frac{a_s^2 d_s + 3b_s^2 c_s}{2a_s b_s d_s}, \tag{22}$$

$\Psi_s(\beta_s)$ is strictly convex on $[1, \beta_{\max}]$. Consequently, the side-wise power-minimizing gain is unique and is obtained by checking the two endpoints and the interior stationary point solving (20), if it lies in $[1, \beta_{\max}]$.

Proof: Under (22), the bracketed term in (21) is non-negative on $[1, \beta_{\max}]$. Since $|\mathcal{A}_s| > 0$ and $\mu_s > 0$, the term $2\mu_s|\mathcal{A}_s|$ yields $\Psi_s''(\beta_s) > 0$ over the interval. Hence Ψ_s is strictly convex and has a unique minimizer. ■

Condition (22) is sufficient but not necessary. When it is violated, $\Psi_s(\beta_s)$ remains continuous on $[1, \beta_{\max}]$, so a bounded one-dimensional search is still well-defined. In the simulations, $\beta_s = 1$ is used for fully passive or zero-active-coherent cases. Otherwise, $\Psi_s(\beta_s)$ is minimized over $[1, \beta_{\max}]$, with (18) used only as an SE-oriented reference or initialization.

C. Extension to TS and SS

Since ES is covered by (14), for $p \in \{\mathrm{TS}, \mathrm{SS}\}$ let $\mathcal{P}_s^{(p)}$ and $\mathcal{A}_s^{(p)}$ denote the passive and active support sets. Define

$$a_s^{(p)} = \min_k \sum_{n\in\mathcal{P}_s^{(p)}} u_{s,k,n},$$
$$b_s^{(p)} = \min_k \sum_{n\in\mathcal{A}_s^{(p)}} u_{s,k,n},$$
$$c_s^{(p)} = \sigma_0^2,$$
$$d_s^{(p)} = \max_k \sigma_a^2 \sum_{n\in\mathcal{A}_s^{(p)}} |h_{s,k,n}|^2.$$

For TS, $\mathcal{P}_s^{(\mathrm{TS})} = \{1, \ldots, N\} \setminus \mathcal{A}_s$ and the effective target becomes R_s/τ_s. Its consumed power is time-averaged:

$$P_{\mathrm{tot}}^{\mathrm{TS}} = P_{\mathrm{ctrl}} + P_{\mathrm{bias}}|\mathcal{A}| + f_u(E_{\mathrm{sw},0} + E_{\mathrm{sw},1}|\mathcal{A}|) + \sum_{s\in\{t,r\}} \tau_s\left(\frac{P_{\mathrm{tx},s}}{\eta_{\mathrm{PA}}} + \mu_s|\mathcal{A}_s|(\beta_s^2 - 1)\right), \tag{23}$$

where $P_{\mathrm{tx},s}$ is the BS transmit power in slot s. For SS, $\mathcal{P}_s^{(\mathrm{SS})} = \mathcal{S}_s \setminus \mathcal{A}_s$, with $\mathcal{S}_t \cup \mathcal{S}_r = \{1, \ldots, N\}$ and $\mathcal{S}_t \cap \mathcal{S}_r = \emptyset$. The ES rules then apply after replacing (a_s, b_s, c_s, d_s) by $(a_s^{(p)}, b_s^{(p)}, c_s^{(p)}, d_s^{(p)})$ and using the corresponding consumed-power denominator.

V. Sparse activation algorithm

For a target sum-SE level ϵ, Algorithm 1 scans the L- and α-grids, forms a compact pool of side-specific active supports, updates β_s, and retains the feasible candidate $(\mathcal{A}_t, \mathcal{A}_r, \beta_t, \beta_r, \alpha)$ with the minimum consumed power.

A. Grid search over sparsity and rate split

The outer grid contains the active-cardinality budget $L \in \mathcal{L}$ and the side-rate split $\alpha \in \mathcal{G}_\alpha$. For each pair (L, α), the side targets are

$$R_t = \alpha\epsilon, \qquad R_r = (1 - \alpha)\epsilon.$$

Thus, L controls hardware sparsity, while α controls traffic asymmetry. Each grid point defines one candidate design subproblem: select L active elements, assign them to sides, update (β_t, β_r), evaluate the constraints of (12), and compare the resulting P_{tot}.

B. Active-set construction

For a fixed (L, α), candidate supports are constructed by testing whether an inactive element $n \notin \mathcal{A}_s$ should be switched on for side s. The test uses the first-order change of the side-wise cost in (19). Define

$$C_s = \frac{K_s(2^{R_s/K_s} - 1)}{\eta_{\rm PA}},$$

which collects the side-rate and PA-efficiency factors in the transmit-power term. Introduce a relaxed activation variable $x_{n,s} \in [0,1]$, with $x_{n,s} = 0$ denoting inactive and $x_{n,s} = 1$ denoting active. Switching on element n changes its coefficient from passive gain 1 to active gain β_s, giving the worst-user coherent increment

$$\Delta u_{n,s} = \sqrt{\rho_s}|g_n| \min_k |h_{s,k,n}|.$$

The coherent denominator is therefore perturbed as

$$D_{n,s}(x_{n,s}) = a_s + b_s\beta_s + (\beta_s - 1)x_{n,s}\Delta u_{n,s}.$$

The active-noise coefficient changes from d_s to $d_s + x_{n,s}\Delta d_{n,s}$. The increment involves the post-insertion maximizing user and is nonsmooth. For ranking and first-order differentiation, the following approximation is used:

$$\Delta d_{n,s} \approx \rho_s \sigma_a^2 \overline{|h_{s,k,n}|^2},$$

where

$$\overline{|h_{s,k,n}|^2} \triangleq \frac{1}{K_s} \sum_{k \in \mathcal{K}_s} |h_{s,k,n}|^2.$$

Substituting these perturbations into the cost in (19) gives

$$\begin{aligned}\Psi_s(x_{n,s}) \approx C_s &\frac{c_s + \beta_s^2(d_s + x_{n,s}\Delta d_{n,s})}{D_{n,s}^2(x_{n,s})} \\ &+ \mu_s(|\mathcal{A}_s| + x_{n,s})(\beta_s^2 - 1) \\ &+ x_{n,s}(P_{\rm bias} + f_u E_{\rm sw,1}).\end{aligned}$$

Differentiating at $x_{n,s} = 0$ gives

$$\begin{aligned}\left.\frac{\partial \Psi_s}{\partial x_{n,s}}\right|_{x_{n,s}=0} &= -\kappa_{1,s}\Delta u_{n,s} + \kappa_{2,s}\Delta d_{n,s} \\ &\quad + P_{\rm bias} + f_u E_{\rm sw,1} + \mu_s(\beta_s^2 - 1),\end{aligned}$$

where

$$\kappa_{1,s} = \frac{2C_s(c_s + d_s\beta_s^2)(\beta_s - 1)}{(a_s + b_s\beta_s)^3}, \qquad \kappa_{2,s} = \frac{C_s\beta_s^2}{(a_s + b_s\beta_s)^2}.$$

Hence, element n is beneficial in the first-order sense when

$$\tilde{\xi}_{n,s} \triangleq \frac{\kappa_{1,s}\Delta u_{n,s}}{\kappa_{2,s}\Delta d_{n,s} + P_{\rm bias} + f_u E_{\rm sw,1} + \mu_s(\beta_s^2 - 1)} > 1.$$

This condition indicates that the transmit-power saving from coherent-gain improvement must exceed the active-noise, bias/switching, and amplifier-power penalties.

For low-complexity ranking, the normalized scores are

$$\begin{aligned}\xi_{n,t} &= \frac{\alpha\sqrt{\rho_t}|g_n| \min_k |h_{t,k,n}|}{\lambda_1\rho_t\sigma_a^2\overline{|h_{t,k,n}|^2} + \lambda_2(P_{\rm bias} + f_u E_{\rm sw,1})}, \\ \xi_{n,r} &= \frac{(1-\alpha)\sqrt{\rho_r}|g_n| \min_k |h_{r,k,n}|}{\lambda_1\rho_r\sigma_a^2\overline{|h_{r,k,n}|^2} + \lambda_2(P_{\rm bias} + f_u E_{\rm sw,1})},\end{aligned} \tag{24}$$

where $\lambda_1 = \sigma_0^{-2}$ and $\lambda_2 = (P_{\max}^{\rm BS})^{-1}$ are normalization weights. The scores in (24) generate candidate supports: for each element, the larger side-wise score determines its assigned side, and the top L elements form one candidate support. A strength-only ranker using the corresponding channel-only numerators in (24) forms a second support and is reported as the "strongest greedy" baseline. After gain update, all candidates are re-evaluated using (5), (9), and (19).

C. Gain update

For each candidate $(\mathcal{A}_t, \mathcal{A}_r)$, the solver computes (a_s, b_s, c_s, d_s) for $s \in \{t, r\}$ and updates β_s on each side. If $|\mathcal{A}_s| = 0$ or $b_s = 0$, no useful active coherent term is available and $\beta_s = 1$ is set. Otherwise, the gain update solves

$$\min_{\beta_s \in [1, \beta_{\max}]} \Psi_s(\beta_s).$$

The updated gains determine the BS power through (9). The candidate is feasible if it satisfies the BS power budget, the bound $1 \le \beta_s \le \beta_{\max}$, non-overlapping active supports ($\mathcal{A}_t \cap \mathcal{A}_r = \emptyset$), protocol-specific ES/TS/SS constraints, and the optional safety limit Γ_s, i.e., the constraints of (12) under the scalar model. Among all feasible candidates for the current ϵ, the one with the smallest $P_{\rm tot}$ is retained as the boundary point.

Algorithm 1 Sparse-Activation Solver

Require: Target ϵ, grids $\mathcal{L}$, $\mathcal{G}_\alpha$, maximum gain $\beta_{\max}$
Ensure: Minimum-power feasible candidate
1: Initialize $P_{\min} \leftarrow +\infty$ and best $\leftarrow \emptyset$.
2: **for all** $(L, \alpha) \in \mathcal{L} \times \mathcal{G}_\alpha$ **do**
3: Build two ranked supports.
4: Update (β_t, β_r), evaluate BS power and $P_{\rm tot}$.
5: Retain any feasible lower-power candidate.
6: **end for**
7: **return** best.

D. Complexity

Let $G = |\mathcal{G}_\alpha|$, $L_g = |\mathcal{L}|$, $K = K_t + K_r$, and Q_β denote the bounded one-dimensional gain-search budget. For each boundary point, score computation, sorting, support evaluation, and gain update require $\mathcal{O}(L_g G(N\log N + NK + Q_\beta))$ operations. For E sampled SE targets, the total complexity is $\mathcal{O}(EL_g G(N\log N + NK + Q_\beta))$, with memory dominated by channel coefficients and element scores.

VI. Numerical results

All numerical settings are listed in Table I. All plotted Monte-Carlo (MC) points average 10000 independent channel realizations with seeds 1–10000. For reproducibility, the source code is available at https://github.com/hjhuang1989-svg/hybrid-star-ris.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Elems./users $(N/K_t/K_r)$	24 / 2 / 2
Nakagami (m_{BR}/m_{RU})	1.0 / 1.0
Avg. powers $(\Omega_{BR}/\Omega_t/\Omega_r)$	0.20 / $(0.060, 0.050)$ / $(0.080, 0.060)$
ES/TS split $(\rho_t, \rho_r)/(\tau_t, \tau_r)$	$(0.55, 0.45)$ / $(0.55, 0.45)$
Noise std. (σ_0/σ_a)	0.25 / 0.05
Caps $(\beta_{\max}/P_{\max}^{\rm BS}$ [W])	5 / 5.0
Ctrl./bias $(P_{\rm ctrl}/P_{\rm bias}$ [W])	0.10 / 0.015
Switching $(f_u/E_{\rm sw,0}/E_{\rm sw,1})$	50 Hz / 2.0×10^{-4} J / 1.2×10^{-4} J
Amp./PA $(\mu_t/\mu_r$ [W]$/\eta_{\rm PA})$	0.0025 / 0.0025 / 0.40
Grids $(\mathcal{G}_\alpha/\mathcal{L})$	$\{0.20, 0.25, \ldots, 0.80\}$ / $\{0, 3, \ldots, 24\}$
Fixed-unif./rand $(L/Q_{\rm rand})$	6 / 1
Safety cap (Γ_s)	inactive

For ES, the baselines are passive ($L = 0$), fixed uniform ($L = 6$ linearly spaced indices $\{1, 5, 10, 14, 19, 24\}$), optimized uniform (linearly spaced support with $L \in \mathcal{L}$), single-random sparse (one random support per (L, α), assigned by the larger score), strongest greedy (same grid with channel-strength ranking only), and active ($L = N$). A stronger $Q_{\text{rand}} = 20$ random-search audit is stored in the reproducibility results and does not change the main ordering at $R_\Sigma = 12$.

Fig. 2 shows the ES EE-SE boundary and the core power-allocation tradeoff. Passive STAR-RIS is competitive at low SE, but its BS-power requirement grows rapidly with R_Σ and becomes infeasible once it exceeds $P_{\max}^{\text{BS}}$. The unfilled passive markers at $R_\Sigma = 18$ and 20 bit/s/Hz are averaged over feasible trials only, with $p_f = 0.98$ and 0.72. The active benchmark lowers BS power but pays large static, switching, and amplification overhead. Compared with strongest greedy, the proposed ranker penalizes active-noise and hardware costs, preserving most coherent gain with lower consumed power. At $R_\Sigma = 12$ bit/s/Hz, the proposed EE is 8.94 bit/J/Hz, improving over fixed uniform, optimized uniform, single-random sparse, active, and passive by 33.1%, 19.4%, 15.2%, 32.4%, and 94.8%, respectively.

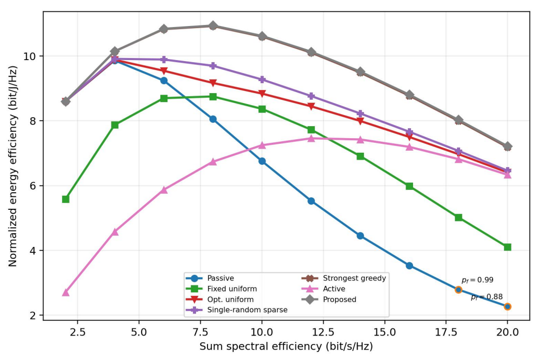

Fig. 2. Approximated ES EE-SE boundary. Unfilled passive markers denote rate points where not all trials are feasible. The label p_f gives the feasible-trial fraction.

Fig. 3 compares ES, TS, and SS in terms of active ratio and common gain. The gains increase with target SE and approach $\beta_{\max}$ when rate pressure dominates. TS requires the densest active set because slot targets inflate by $1/\tau_s$, whereas ES and SS maintain lower active ratios with comparable gains. At $R_\Sigma = 12$ bit/s/Hz, the ES/TS/SS active ratios are about $0.29/0.53/0.31$, with average gains $4.70/4.96/4.64$.

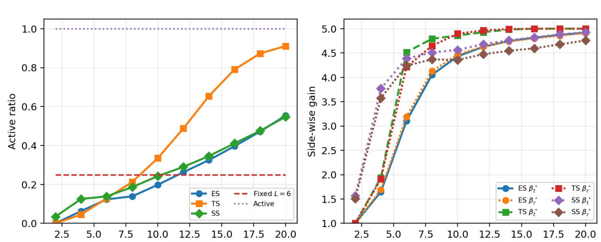

Fig. 3. Cross-mode ES/TS/SS design rules: active ratio and side-wise common gain.

Fig. 4 shows ES hardware sensitivity at $R_\Sigma = 6$, 9, 12, 15, and 18 bit/s/Hz. A larger gain is not always beneficial: increasing σ_a suppresses the selected gain because amplifier noise scales with β_s^2 and raises the required power. A larger f_u likewise increases switching power and reduces EE.

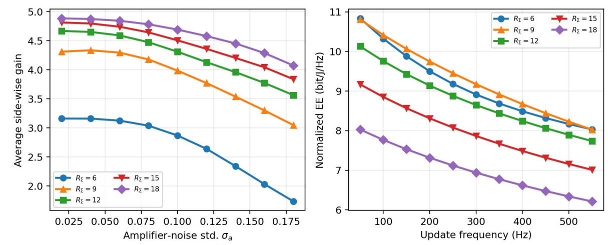

Fig. 4. ES hardware sensitivity versus amplifier noise and switching overhead.

Fig. 5 compares ES power at $R_\Sigma = 12$ and 16 bit/s/Hz. At $R_\Sigma = 12$, active operation minimizes BS power (0.67 W) but incurs large overhead, whereas passive operation needs 2.64 W. The proposed sparse design achieves the lowest total power, about 1.37 W, and the same trend persists at $R_\Sigma = 16$.

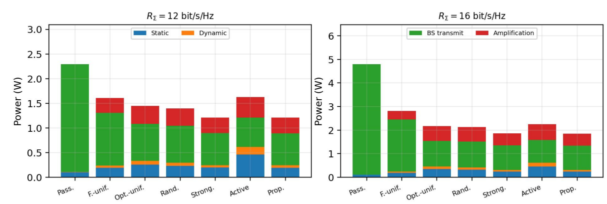

Fig. 5. ES power breakdown at $R_\Sigma = 12$ and 16 bit/s/Hz.

VII. Conclusion

This letter characterized the EE-SE tradeoff of hybrid active-passive STAR-RIS with sparse active amplification and gain control. An amplification on–off criterion, an SE-oriented gain law, and a power-minimizing gain refinement were embedded into a finite-grid sparse-activation solver. Numerical results show that sparse hybrid activation avoids both passive BS-power burden and full-array active overhead. Future work will validate the design under full-dimensional beamforming and incorporate signal-dependent active-output constraints.

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